

WHEN DOES JOINT KEY-CACHE ALLOCATION PAY? OUTPUT-DISTORTION REGIMES AND THEIR LIMITS ON END TASKS

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ABSTRACT

A KV-cache budget can be spent by quantizing every token, by evicting tokens, or by mixing zero and finite precisions across tokens. Hybrid systems already exist; we ask *when* the mixed interior improves over its two endpoints, and whether the answer measured on attention-output distortion carries over to end tasks. Under a separable local surrogate, key-token precision becomes a discrete rate-allocation problem that contains eviction as a zero-rate tier, and a finite-bit tier is dominated when its absolute logit-noise cost exceeds that of zero rate. The fraction of heads with a dominated 2-bit tier orders the interior’s exactly recomputed output-error advantage across six LLMs and 25 model–context cells (Spearman $\rho = -0.978$; -0.94 controlling for model), but the crossing point is architecture-specific, and longer contexts move a fixed model toward the eviction corner. Over 4,096 decode tokens, a head’s best policy family is stable while its token allocation goes stale within steps. An end-task audit against quantization and eviction baselines then bounds these findings. The output-error-optimal allocator is the strongest *token-selective* policy we test (0.77 vs. 0.60 mean task score), but on both end-task models a dense low-bit quantizer beats every token-selective policy once the quantizer suits the model: TurboQuant on Llama-3.1-8B (0.99 vs. 0.65) and per-channel KIVI or KVQuant on Qwen3-8B, where TurboQuant itself loses (8k: 0.99 vs. 0.95 for the allocator and 0.90 for TurboQuant). Output distortion predicts accuracy only weakly across methods and misprices rotated quantization noise. The allocator’s dominant failure is answer truncation, which smoothing its token scores over neighboring positions largely removes (8k: 39% to 9% of answers).

1 INTRODUCTION

At long context, the KV cache can bound batch size and decode throughput. A cache budget admits at least three actions: assign every token one low precision (Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2025), retain selected tokens at a higher precision (Zhang et al., 2023; Liu et al., 2023; Li et al., 2024; Xiao et al., 2024), or mix zero and finite rates across tokens, the *interior* between those two endpoints. The last option is no longer hypothetical: DiffKV and HqeKV combine pruning with multiple precision levels (Zhang et al., 2025b; Wang et al., 2026). What remains unclear is whether the extra interior choices are useful for a given model and deployment context, whether a simpler endpoint already suffices, and whether the proxy used to design such allocators predicts what a user observes.

This paper is a measurement study of those questions. It does not propose a better compressor. We build the allocator that is optimal for attention-output distortion, SIEVE, and use it as an instrument: it tells us how large the interior’s advantage can be under that objective, where the advantage disappears, and whether it survives the step to task accuracy. The motivating reversal is within one model: from 8k to 128k, the share of Llama-3.3-70B’s heads whose 2-bit tier is dominated by eviction (D_2 , §2) rises from 5.5% to 41.4%, the interior’s hindsight portfolio gain (G_+ , §3) falls from $3.17\times$ to $1.40\times$, and the share of heads where the interior beats both endpoints by $2\times$ (the *band*) falls from 87.4% to 21.3%. A verdict at one context therefore need not transfer to another.

Table 1: Summary of findings, their evidence, and what they do *not* show.

	Observation	Evidence	Boundary
F1	Tier extinction (D_2) orders the interior’s output-error advantage	25 cells, $\rho = -0.978$; Fig. 1a	crossing is per architecture, not universal
F2	Longer context moves a fixed model toward the eviction corner	6 trajectories, 4k–256k; Fig. 1b	mostly absolute length, small RoPE-cap residual
F3	Policy family is slow; token allocation is fast	6 cells, 4,096 tokens; Fig. 2b	output error only
F4	The output-error optimum is the strongest token-selective policy, but a model-matched dense quantizer beats all of them	36 task cells (7 methods) + 14 at 8k (all 13); Table 2, Fig. 3a	best quantizer differs by model; 32k/128k rerun pending
F5	Output distortion ranks budgets and contexts well and methods weakly	within-cell $\rho = -0.36$ (8k, 11 methods); Fig. 3b	misprices rotated noise; full grid pending
F6	Token-granular allocation truncates multi-token answers; positional smoothing largely repairs it	36% vs. 1–11% (128k); 39% $\rightarrow$ 9% (8k); Fig. 3c	pre-registered 128k test pending
F7	The comparison set and information contract move verdicts more than prompt sampling does	4 contrasts; Table 3	4 cells for the sampling estimate

Findings. We organize the paper around seven observations (Table 1). F1–F3 concern attention-output distortion, where the interior is well defined and exactly measurable (§4). F4–F6 test whether those regimes transfer to retrieval accuracy against published quantization and eviction baselines (§5); the answer is mixed, and the way it fails is informative. F7 cuts across both levels: the comparison itself moves verdicts more than sampling does.

Contributions.

1. **A common action space and a measurable coordinate.** We place within-head key-token precision and eviction under one local output-distortion surrogate, keep the exact set-eviction error distinct from its separable zero-rate approximation, and derive a tier-extinction coordinate D_2 (§2).
2. **A regime map for output distortion.** With exact recomputation and a symmetric information contract, we show where the interior pays, how context moves a model across regimes, and on which two timescales a controller would have to act (F1–F3).
3. **An end-task audit of the map.** Against quantization and eviction baselines at matched key bits, we identify where output-distortion optimality transfers to task accuracy, where it does not, and the mechanism of the largest failure (F4–F6).

Concurrent RDKV also derives joint eviction–quantization rate allocation (Zhang et al., 2026b). Our contribution is complementary: we measure *when* a token-precision interior lowers attention-output error, how that changes with context, and how far that proxy carries to task accuracy. Section 7 compares with RDKV and other allocation axes.

2 JOINT TOKEN-RATE ALLOCATION AS A COMMON FOUNDATION

Scope and local quantization surrogate. For one query q , keys k_i , and values v_i , let $s_i = q^\top k_i / \sqrt{d}$, $a = \text{softmax}(s)$, and $o = \sum_i a_i v_i$. Quantizing key i perturbs its logit by δ_i , and the first-order output change is $\Delta o \approx \sum_i a_i (v_i - o) \delta_i$. If residual perturbations are zero mean and cross-token covariance terms vanish,

$$\mathbb{E} \|\Delta o\|^2 \approx \sum_i w_i d_{b_i}, \quad w_i = a_i^2 \|v_i - o\|^2, \quad d_b = \text{Var}(\delta_b). \quad (1)$$

Only a common-mode logit shift is canceled exactly by softmax; token-dependent bias and covariance are outside this surrogate. We therefore use Eq. 1 to *choose* allocations, never to score their reported error.

Exact eviction versus a zero-rate surrogate. For an evicted set E with mass $A_E = \sum_{i \in E} a_i$, renormalizing the surviving attention gives exactly

$$\|o_{\setminus E} - o\|^2 = \frac{\|\sum_{i \in E} a_i (v_i - o)\|^2}{(1 - A_E)^2}. \quad (2)$$

The denominator and cross-token inner products make set eviction nonseparable; for a singleton j the magnitude is $a_j \|v_j - o\| / (1 - a_j)$, as derived by CAOTE (Goel et al., 2025). To include zero rate in a token-separable allocator, we use the small-evicted-mass, cross-term-free approximation $\tilde{D}_0(E) = \sum_{i \in E} w_i$ and normalize its *absolute* cost to $d_0 = 1$. Eviction is therefore a zero-rate action *inside the surrogate*, not an exact independent-token equality.

The implemented discrete allocation. For tiers $\mathcal{B} = \{0, 1, 2, 3, 4, 5, 6, 8\}$ and an average key budget B , we solve

$$\min_{b_i \in \mathcal{B}} \sum_i w_i d_{b_i} \quad \text{s.t.} \quad \sum_i b_i \leq BL, \quad b_i^*(\lambda) = \arg \min_{b \in \mathcal{B}} \{w_i d_b + \lambda b\}, \quad (3)$$

with a bisection on λ enforcing the budget. Relaxing bits to be continuous with $d_b \propto 4^{-b}$ gives $b_i^* = \log_2 a_i + \log_2 \|v_i - o\| + c$, which explains the reverse-water-filling shape; Eq. 3 is the optimizer used in all experiments.

Two coordinates, not one universal phase boundary. If $d_b \geq d_0$, tier b consumes bits without lowering the surrogate cost and is dominated. This condition is sufficient for extinction but not necessary. We summarize the sharp side by

$$D_2 = H^{-1} \sum_{h=1}^H \mathbb{I}[d_{h,2} \geq d_0], \quad (4)$$

the fraction of heads in a model–context cell whose 2-bit tier is dominated. The diffuse side is different: unequal allocation has little leverage when token importance varies less than the tier spacing. For attention-only importance, $\text{std}_i(\log_2 a_i) = \tau / \ln 2$ is an algebraic identity, not evidence of optimality. Thus D_2 measures approach to the eviction-like corner, and a dispersion coordinate is needed for the uniform corner. D_2 contains no explicit L , but it is measured *at* a context and changes with it.

3 MEASUREMENT PROTOCOL

We measure at two levels. Attention-output distortion is exact and cheap, so it supports a dense regime map over six models and 4k–256k tokens (§4). End-task accuracy is the quantity users observe but is expensive, so we audit the map on two models chosen to lie at opposite ends of it, with a third pending (§5).

Output-distortion level: models, cells, and units. We evaluate Llama-3.1-8B and Llama-3.3-70B (Grattafiori et al., 2024), Mistral-7B (Jiang et al., 2023), Qwen1.5-MoE-A2.7B (Qwen Team, 2024), Qwen3-8B, and Qwen3-30B-A3B (Yang et al., 2025). A symmetric grid has 16 model–context cells between 8k and 128k, and its extension has 25 cells spanning 4k–256k (44,416 query-head–configuration aggregates). Across the six architectures there are 10,240 query-head and 2,016 KV-head positions, reused across contexts, so statistical claims treat cells and models, not heads, as independent units. PG-19 text (Rae et al., 2020) supplies the haystack, and every cell passes a full-cache needle-retrieval gate.

Compression, budget, and information contract. Keys are quantized with TurboQuant_{mse} (Zandieh et al., 2025) (random rotation and Lloyd–Max scalar quantization (Lloyd, 1982)). The headline budget is $B = 3$ key bits per context token; tier 0 evicts and tier 8 stores an 8-bit key. Values remain uncompressed. The primary comparison is symmetric: the mixed allocator and the eviction endpoint both use only lagged attention, and positions unseen by that score stay at the maximum tier. The endpoints, which we also call *corners*, are uniform 3-bit quantization and accumulated-attention eviction keeping the number of 8-bit keys the budget allows (*uniform+accum*); a robustness analysis takes the minimum over five corners. Oracle results are upper bounds only.

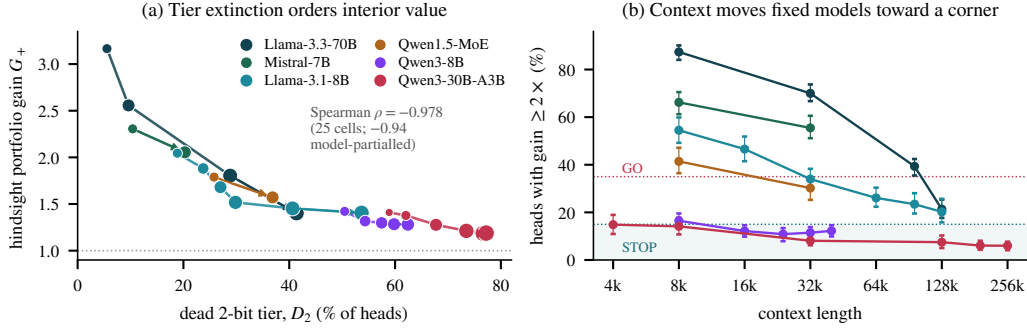


Figure 1: **F1–F2: where the interior lowers output error.** Mixed allocator and eviction endpoint both use lagged information; exact recomputation scores both. **(a)** The tier-extinction coordinate D_2 orders hindsight portfolio gain G_+ across 25 cells; arrows show increasing context. **(b)** Context moves each fixed model across predeclared output-error bands. Error bars are 90% layer-cluster intervals. Thresholds are not universal across architectures.

Exact response and metrics. For every assignment we quantize the real keys, recompute attention, and measure the output error exactly. For head h ,

$$g_h = \frac{\min(D_h^{\text{uniform}}, D_h^{\text{evict}})}{D_h^{\text{mixed}}}, \quad G_+ = \exp[H^{-1} \sum_h \log \max(g_h, 1)]. \quad (5)$$

G_+ , the primary cell response, is a hindsight portfolio statistic, not achieved router performance. The share of heads with $g_h \geq 2$ (the band) is a secondary summary, with predeclared thresholds: a cell is GO when its band is at least 35%, STOP below 15%, and NARROW in between.

End-task level. Llama-3.1-8B (8k, 32k, 128k) and Qwen3-8B (8k, 32k) share size and GQA ratio but sit at opposite ends of the map (low vs. high D_2). Four RULER-style tasks (Hsieh et al., 2024) (single-needle, multi-key, multi-value retrieval, and variable tracking) use 20 held-out prompts per cell. The cache is compressed once, after the context is prefilled and *before* the question arrives, as in prefix caching; compressing after the question lets observation-window scores read it and saturates every such method at 1.00. Budgets are $B \in \{2, 3\}$ key *code* bits per context token, audited at run time; values, the last 32 context tokens, and generated tokens are uncompressed in every method. Side information (norms, scales, outlier indices, width maps) is reported as effective bits. Blocks where the uncompressed model scores below 0.9 are excluded (only Qwen3-8B multi-value), leaving 36 valid model/context/task/budget cells. Mistral-7B (8k, 32k), which shares the GQA ratio and sits on Llama’s side of the map, is being added as a third model under the same protocol.

Baselines and the instrument. Quantization: TurboQuant-MSE, KIVI (Liu et al., 2024), and KVQuant (Hooper et al., 2024). Eviction, all keeping $\lfloor BC/8 \rfloor$ 8-bit keys: H2O (Zhang et al., 2023), SnapKV (Li et al., 2024), DropKV (Zhang et al., 2026a), Ada-KV (Feng et al., 2025), LaProx (Mai & Kim, 2026), and OBCache-K (Gu et al., 2026) with Ada-KV allocation. SIEVE is Eq. 3 per KV head with a 4-bit cascade score (§4), plus a router that assigns each KV head to the mixed interior, uniform quantization, or eviction by minimum output error on 10 disjoint calibration prompts. SIEVE’s router is therefore *output-error-optimal by construction*; its end-task behavior tests the proxy, not a tuned method. Implementation details and deviations are in Appendix H.

4 OUTPUT-DISTORTION REGIMES

F1: Tier extinction orders the interior’s value, but the crossing is per architecture. On the 16-cell grid, demoting both the mixed allocator and the eviction endpoint to lagged information retains roughly half of the apparent asymmetric advantage (Fig. 2a): the symmetric band stays 1.2–12.5 points above the all-oracle band in all 16 cells, five cells fall below 15% (STOP), and G_+ ranges from $1.05\times$ to $3.21\times$. D_2 still orders the symmetric band at $\rho = -0.985$. Extending to 25 cells, D_2 has Spearman $\rho = -0.978$ with both G_+ and the band, and -0.94 with G_+ (-0.915 with the band) after

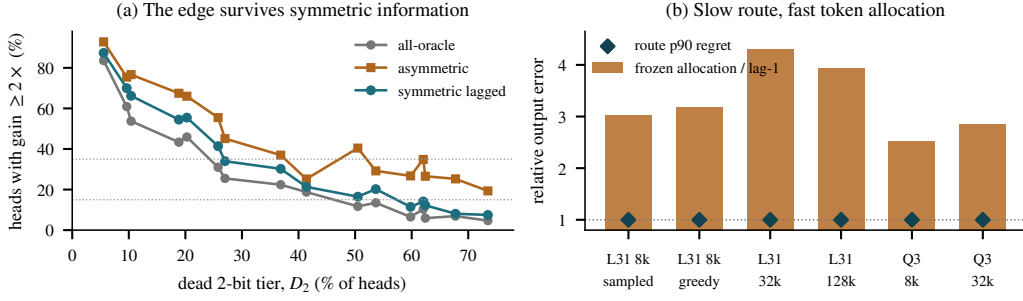


Figure 2: **F1 robustness and F3 timescales.** (a) On the same 16 cells, the lagged-versus-lagged advantage is smaller than the asymmetric estimate (current-query interior against lagged eviction) but remains ordered by D_2 . (b) In six cells, keeping the calibration-time *policy family* has p90 regret 1.00 through 4,096 generated tokens, whereas freezing the token assignment costs $2.52\text{--}4.30\times$ a one-step-old assignment.

partialling out model identity (Fig. 1a). The crossing, however, is not universal. Pooled monotone crossings sit near $D_2 = 29.0\%$ (GO) and 54.1% (STOP), yet Llama-3.1-8B holds 20.2% band at $D_2 = 53.7\%$ while Qwen3-8B is at 12.2% at $D_2 = 54.3\%$, an eight-point gap across 0.6 points of D_2 that exceeds prompt uncertainty. D_2 is an ordering coordinate; any operational boundary must be fitted per architecture. The location also depends on what information each side of the comparison receives: on the same 16 cells, STOP is crossed at $D_2 = 46.9\%$ when both sides see the current query and at 57.4% when both use lagged attention, a 10.5-point shift larger than either confidence interval, and under the asymmetric contract it is never crossed. D_2 itself does not share this sensitivity: across two campaigns with different corner sets and code it agrees to within 0.30 points on all 15 shared cells. How much the band depends on the comparison is taken up in F7.

Why D_2 rather than a concentration or context statistic? We compared D_2 with simpler predictors on the same 25 cells, under a protocol fixed before computing it (Table 6): attention concentration τ , the spread of log token sensitivity (a heterogeneity statistic of the kind RateQuant uses (Zuo et al., 2026)), attention entropy, and context length alone. Within a model, every statistic that moves with context orders G_+ almost perfectly (mean within-model $\rho = -0.99$ for D_2 , τ , the heterogeneity statistic, and context length). The difference is across architectures. Pooled over models, D_2 reaches $\rho = -0.978$, against -0.72 for τ and heterogeneity and -0.48 for context alone, and a log-linear fit on five models predicts the sixth to within a factor of 1.12 for D_2 , against 1.15 and 1.24. D_2 is therefore most useful for comparing architectures, and its held-out advantage over dispersion statistics is modest.

F2: Context moves a fixed model toward the eviction corner. Every measured trajectory moves toward larger tier extinction with context (Fig. 1b). Llama-3.3-70B goes from GO at 8k to NARROW at 128k; Qwen3-8B goes from NARROW at 8k to STOP by 16k. The mechanism is mostly absolute length: attention concentration τ is generally convex in $\log L$. Reaching the trained RoPE cap adds a residual of about $+0.21$ in Qwen3-8B and Qwen3-30B, but none in Llama-3.1-8B and none at 0.75 of the cap. Both context and architecture matter; the data do not support a universal RoPE-triggered transition. The surrogate is useful but not exact: predicted and recomputed gains agree to a median ratio in $[0.79, 1.34]$ (a fit check, not a held-out prediction).

F3: Two timescales, and the cost of acting on them. A head’s output-error-optimal family (interior, uniform, or eviction) is stable over 4,096 generated tokens in all six measured cells, while the token allocation inside a mixed head goes stale within steps (Fig. 2b). Any controller built on this map must therefore route rarely but re-budget often. Table 5 (App. F) prices the two constraints such a controller meets: sharing one allocation per GQA group costs $1.14\times$ at $n_{\text{rep}} = 4$, and a 4-bit cascade recovers 89% of the staleness penalty.

Table 2: **End-task audit:** mean RULER score over valid cells (FP = 0.999). *Full grid:* 36 cells, one run each. *8k rerun:* all methods in one process on the same prompts (measured). Output error and effective bits ($B = 2$) from the 8k rerun. *Hard:* Llama-3.1-8B, 32k, 32-key, $B = 2$. **Blue[†]:** estimates awaiting runs (App. I).

family	method	full grid (36 cells)			8k rerun (14 cells)				
		Llama	Qwen	all	Llama	Qwen	out. err.	eff. bits	hard
quant.	TurboQuant-MSE	0.985	0.867	0.946	0.997	0.895	0.439	2.12	0.88[†]
	KIVI ($g=128$)	0.96[†]	0.97[†]	0.96[†]	0.964	0.987	0.300	2.25	0.85[†]
	KIVI ($g=32$)	0.98[†]	0.99[†]	0.98[†]	0.986	1.000	0.203	3.00	0.88[†]
	KVQuant	0.97[†]	0.98[†]	0.97[†]	0.976	0.990	0.163	2.33	0.88[†]
eviction	H2O	0.38[†]	0.33[†]	0.36[†]	0.354	0.358	0.341	2.04	0.25[†]
	SnapKV	0.479	0.237	0.398	0.348	0.203	0.324	2.04	0.21[†]
	DropKV	0.459	0.334	0.418	0.331	0.262	0.345	2.04	0.18[†]
	Ada-KV	0.612	0.285	0.503	0.464	0.233	0.326	2.04	0.25[†]
	LaProx	0.640	0.457	0.579	0.540	0.455	0.330	2.04	0.32[†]
	OBCache-K + Ada-KV	0.719	0.369	0.602	0.580	0.298	0.334	2.04	0.39[†]
mixed	SIEVE interior only	0.479	0.438	0.465	0.467	0.508	0.165	2.07	0.28[†]
	SIEVE (router)	0.654	0.989	0.766	0.680	0.953	0.150	2.08	0.32[†]
diag.	SIEVE, pooled score	—	—	—	0.779	0.633	0.196	2.07	—
	SIEVE, oracle router	—	—	—	0.834	0.967	0.136	2.08	—

5 DO OUTPUT-DISTORTION REGIMES TRANSFER TO END TASKS?

F4: The output-error optimum is the strongest token-selective policy, but a model-matched dense quantizer beats all of them. Table 2 and Fig. 3a show three regularities. First, among policies that choose *which tokens* to keep, SIEVE scores highest: 0.766 against 0.602 for OBCache-K + Ada-KV over the full grid (+0.16, 90% CI [+0.13, +0.20] by prompt bootstrap), and in the 8k rerun it beats every eviction method, including H2O, on average (0.797 against at most 0.504) with less than half their output error. Second, on Llama-3.1-8B, keeping every token at low precision dominates: TurboQuant scores 0.985, and at $B = 2$ every token-selective method trails it by at least 0.35 (best: OBCache-K + Ada-KV, 0.626 vs. 0.976; SIEVE, 0.429). Third, on Qwen3-8B, TurboQuant’s per-token rotated 2-bit keys break down (variable tracking scores 0.46 at 8k and 0.40 at 32k) and SIEVE beats it (0.989 vs. 0.867). That reversal belongs to the quantizer, not to dense quantization. In the 8k rerun the per-channel quantizers recover Qwen: KIVI scores 0.987 at $g=128$ and 1.000 at $g=32$, and KVQuant 0.990, all above SIEVE (0.953), with KIVI-128 and KVQuant using only 0.17–0.25 more effective bits. On Llama the order of the quantizers reverses (TurboQuant 0.997, KIVI-128 0.964). So on both models some dense quantizer beats every token-selective policy, but which one depends on the model. Mistral-7B looks like Llama on its calibration prompts (TurboQuant 0.93–1.00, SIEVE’s interior 0.00–0.27; router pending, **0.55[†]**). The preregistered correlation between the output-error band and the router’s accuracy gain is -0.70 , opposite in sign to the prediction. All 15 full-grid cells where SIEVE loses to an eviction method are on Llama: 7 of 8 at $B = 2$ below 128k, and all 8 at 128k (Table 9).

F5: Output distortion ranks budgets and contexts well and methods weakly. SIEVE minimizes the distortion it targets and has among the lowest head-output errors (0.120 on the full grid, 0.150 in the 8k rerun), yet dense quantizers beat it. Fig. 3b separates the two uses of the proxy. Across budgets and contexts, error and accuracy move together ($\rho = -0.69$ within the eviction family on the full grid). Across methods at a fixed budget the relation is weak and depends on which methods are compared. With the seven full-grid methods, the mean within-cell rank correlation is -0.05 ; with all eleven methods of the 8k rerun it is -0.36 (-0.46 among token-selective policies, and 0.01 among the four dense quantizers). The exception that breaks the proxy is TurboQuant: its rotated, spread-out noise gives the highest output error (0.44) at the highest Llama accuracy, while per-channel quantizers have low error and high accuracy. Output distortion therefore sizes budgets but misprices at least one quantizer family, and it should not choose between families. The router’s calibration costs part of its accuracy but is not the main culprit: routing each head on its *measured* test-prompt error raises the 8k

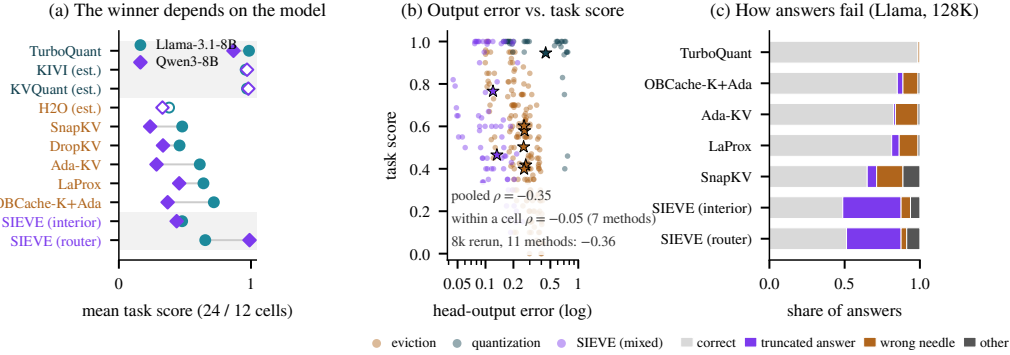


Figure 3: **F4–F6: what transfers to task accuracy.** (a) Mean task score per method and model; hollow markers are estimates awaiting runs (App. I). Full-grid means; KIVI, KVQuant and H2O are full-grid estimates guided by the 8k rerun (Table 2). TurboQuant wins on Llama-3.1-8B; on Qwen3-8B SIEVE beats TurboQuant but not the per-channel quantizers. (b) Head-output error vs. task score for every full-grid (cell, method); stars are method means. Error tracks accuracy across budgets and contexts; across methods it is weak, and TurboQuant (dark, top right) combines the highest error with the highest accuracy. (c) Single- and multi-key answers at Llama-3.1-8B, 128k, $B \in \{2, 3\}$: SIEVE fails by *truncating* the correct number, eviction methods by returning a wrong needle.

score from 0.797 to 0.891 (Llama: 0.680 to 0.834), still below every dense quantizer. The allocation theorem holds for its own objective; the transfer from that objective to retrieval is what fails.

F6: Token-granular allocation truncates multi-token answers. SIEVE’s weakest setting is Llama-3.1-8B at 128k, where it trails OBCache-K + Ada-KV by 0.32 at $B = 2$ (0.442 vs. 0.762). Its wrong answers have a distinctive form (Fig. 3c). Asked for 4929054 it answers “492”; for 9465170, “946”; for 6022964, “602296”. Llama tokenizes digits in groups of three, so the allocator keeps the key and the first token of the value and loses the tokens after it. Truncation accounts for 36% of SIEVE’s single- and multi-key answers at 128k (39% for the interior alone) against 1–11% for every eviction method, which instead fail by returning a different needle (10–18%). Truncation is not specific to 128k: pooled over Llama’s three contexts, 40% of the interior’s answers are truncated. The router does not rescue these heads, because by its objective the interior is the right choice: at 128k it sends 99% of Llama’s KV heads to the interior, whose output error (0.094) is half that of any eviction method. Losing the second token of one needle’s value barely moves an error averaged over heads and answer steps, but changes the answer completely.

Why do the eviction methods avoid truncation? Every one of them that beats SIEVE here smooths its token score over neighboring positions before selecting (max- or average-pooling over 7 positions), so a kept token protects its neighbors and a needle survives as a phrase. Pooling has its own cost: with several targets it can merge evidence that belongs to different ones (Zhang et al., 2026c). SIEVE allocates each token independently from unpooled attention. We pre-registered a test of this explanation: the same water-filling allocator on pooled attention supports it if it recovers at least half of the 128k, $B = 2$ gap to OBCache-K + Ada-KV, and refutes it below one fifth. The 8k rerun already points strongly in that direction. On Llama-3.1-8B single- and multi-key retrieval, pooling cuts truncated answers from 39% to 9% and raises the interior’s score from 0.467 to 0.779, above OBCache-K + Ada-KV (0.580) and the calibrated router (0.680); at $B = 2$ it moves the interior from 0.299 to 0.663, past OBCache’s 0.446. On Qwen3-8B pooling helps less (0.508 to 0.633; router 0.953). The pre-registered 128k cell remains the test (estimate 0.58[†]; half-gap threshold 0.59; oracle router 0.55[†]).

F7: Verdicts depend more on the comparison than on the sample. Across both levels, the largest changes in a verdict came from what a method was compared against, and under which information contract, rather than from which prompts were drawn (Table 3). A new prompt block moves the band by a standard deviation of only 0.4–3.0 points, so six prompts per cell suffice. Each of the four other choices in Table 3 moves a result by more than that and changes at least one verdict: the

Table 3: **F7**: how much each choice moves a verdict, compared with prompt sampling.

Choice varied	Effect	Verdicts changed
Prompt block (sampling)	band s.d. 0.4–3.0 pts, 4 cells	none
Eviction corner: accum vs. best of 5	band −5.1 to −8.6 pts	3 of 4 borderline cells
Information: current query vs. lagged	STOP crossing 46.9% → 57.4% D_2	boundary location
Question visible at compression	SnapKV 0.11–0.45 → 1.00 in 4 cells (same prompts); H2O ≤ 0.07	every eviction ranking
Dense baseline: TurboQuant vs. per-channel	Qwen3-8B 8k: 0.90 → 0.99–1.00	SIEVE vs. dense on Qwen

eviction corner, the information each side sees (F1), whether the question is visible when the cache is compressed, and which dense quantizer stands for the endpoint (F4). A KV-compression verdict is therefore not a measurement until it names its comparison set and information contract.

6 IMPLICATIONS

Five practical conclusions follow, each bounded by the retrieval tasks and models measured so far.

- 1. Report dense low-bit quantization, with more than one quantizer, next to any token-selective method.** On both end-task models a dense quantizer beats every eviction and mixed policy we test at 2–3 key bits, but a different one on each: TurboQuant on Llama-3.1-8B, per-channel KIVI and KVQuant on Qwen3-8B, where TurboQuant loses to SIEVE.
- 2. Use output distortion to size budgets, not to choose methods.** It orders settings well (F1, F5) and methods weakly, and misprices rotated quantization noise (F5). Routing between method families needs a task-aware signal.
- 3. Allocate spans, not tokens.** Token-separable objectives cannot see that a multi-token answer is only useful whole (F6). Smoothing the allocator’s scores over neighboring positions, which leaves Eq. 3 intact, already cuts truncation from 39% to 9% at 8k.
- 4. Name the comparison set and information contract with every verdict.** They move results more than the prompt sample does (F7).
- 5. Calibrate per architecture and per context, and re-budget often.** Crossings are architecture-specific and move with context (F1, F2); allocations go stale within steps (F3).

7 RELATED WORK

Dense KV quantization. Dense quantizers store every token at low precision: per-channel or per-token integer schemes (Liu et al., 2024; Hooper et al., 2024), low-rank and sparse residual correction (Kang et al., 2024), rotations and sketches (Zandieh et al., 2024; 2025), and importance-aware mixed precision (Yang et al., 2024; He et al., 2024). RateQuant allocates finite precision across K/V heads from fitted distortion curves (Zuo et al., 2026). Block-scaled 4-bit formats now make such caches native to hardware: the MX formats (Open Compute Project, 2023) and NVFP4, whose 16-element blocks carry FP8 scales (Alvarez et al., 2025a), including an NVFP4 KV cache (Alvarez et al., 2025b). As F4 shows, this endpoint’s strength depends on the quantizer, so a study of the interior has to report more than one.

Eviction and sparse attention. Evictors keep recent and sink tokens (Xiao et al., 2024), accumulate attention (Zhang et al., 2023; Liu et al., 2023), use the last query (Oren et al., 2024) or a prefill observation window (Li et al., 2024), and split budgets across layers and heads (Cai et al., 2024; Feng et al., 2025). Output-aware scores estimate the attention-output perturbation of removing a token: exactly for a singleton (Goel et al., 2025), under perturbation constraints (Feng et al., 2026), by decoupled residual-output perturbation (Zhang et al., 2026a), by Optimal-Brain-Damage saliency (Gu et al., 2026), through the value and output projection with a model-wide budget (Mai & Kim, 2026), or by layer-wise reconstruction with spatial-temporal smoothing (An et al., 2026). Query-aware sparsity keeps the whole cache and selects pages per query (Tang et al., 2024), and reasoning-oriented methods target long generated traces (Cai et al., 2025; Mao et al., 2026). We evaluate six of these

families at matched key bits. Our truncation finding (F6) suggests one reason the strongest ones work: most smooth their scores over neighboring positions, which protects multi-token spans in a way a token-separable output-error objective does not. We do not claim first output-aware eviction; we state the separable approximation needed to put zero rate inside a finite-rate allocator.

Hybrid storage and rate allocation. The split between quantization and eviction is not strict. DiffKV routes tokens among high precision, low precision, and pruning with an on-GPU manager (Zhang et al., 2025b), and HqeKV optimizes among full precision, several quantized levels, and eviction using joint K–V importance (Wang et al., 2026). Concurrent RDKV assigns zero-to-finite rates to value tokens and key channels with packed decode (Zhang et al., 2026b), and concurrent attention-aware transform coding applies output-aware reverse water-filling to transformed K/V channels (Laus et al., 2026). Our allocation unit is the within-head key token, and our question is not whether water-filling can be derived but when its interior pays and whether the answer survives the step to task accuracy. Head, channel, and token axes are potentially composable (Table 12, App. J).

Diagnostics, evaluation contracts, and systems. Long-context compression is usually evaluated on synthetic retrieval (Hsieh et al., 2024) and on natural long-context tasks (Bai et al., 2025). Closest in spirit to our study, Zhang et al. (2026c) hold an eviction selector’s contract fixed (attention tensor, observation window, budget, projection rule) and vary one stage at a time, attributing failures to a mismatch between prefill and decode queries, to output-unaware scoring, or to a projection that fragments evidence across the budget, on the same three model families we use. Their study covers eviction alone at 5–10% retention and finds that output-aware scoring helps in most cells where it has room to. Ours adds dense and mixed-precision policies and finds that output error fails at a different point: choosing *between* families at a fixed budget (F5). Our truncation failure (F6) is an instance of their projection stage, produced by a mixed-precision allocator rather than a selector, and F7 carries their fixed-contract principle one level up, to the baselines and information contract a verdict is stated under. Deployed eviction also meets infrastructure constraints that our simulated study does not address: fused attention kernels never materialize the attention scores that score-based evictors read (Dao, 2023; Shah et al., 2024; Zhang et al., 2024b), and paged allocators reclaim only empty blocks (Kwon et al., 2023), so scattered survivors free little memory unless the cache is compacted (Mao et al., 2026).

8 LIMITATIONS AND CONCLUSION

The evidence is deliberately scoped. End-task results cover two 8B models (a third, Mistral-7B, is pending) and four synthetic retrieval tasks, so F4–F6 may not hold for reasoning, summarization, or long generation, where a truncated span may matter less or more. Compression is key-only with exact values, budgets match key code bits rather than total KV bytes, and all quantization is simulated, so no memory or throughput claim is made. The regime crossing is architecture-conditioned, and the diffuse (uniform) side of the map has less support than tier extinction. Several end-task entries are estimates pending the runs listed in Appendix I.

Within that scope, joint allocation is not uniformly useful. Tier extinction orders when the interior lowers output error, context moves a model toward a corner, and allocations go stale faster than routes. On end tasks the output-error optimum chooses *which tokens* to keep better than any evictor we test, but a model-matched dense quantizer does better still, and output distortion cannot make that choice. Its truncation failure points to the unit of allocation, which positional smoothing begins to repair.

AI USE STATEMENT

In this work, we used generative AI tools (an AI coding and writing assistant) for the following tasks that the ICLR policy requires us to disclose. *Method implementation*: implementing and testing the KIVI and KVQuant-style baseline quantizers, the end-task result readers, the predictor comparison, and the figure and table scripts. *Experiment-design feedback*: proposing parts of the end-task campaign, namely the harder 32-key operating point, the question-visible control, the third end-task model, the pooled-score and oracle diagnostic arms, the predictor-comparison protocol, and several of their pre-registered decision rules, all of which the authors reviewed before any run. *Hypothesis proposal and results interpretation*: proposing the positional-pooling explanation of answer truncation

(F6), the cross-cutting observation on comparison contracts (F7), and interpretations of the end-task results. *Qualitative data analysis*: classifying the models’ wrong answers as truncated, wrong-needle, or other by rule, from generated text. We also used AI assistance, as the policy recommends disclosing, for editing code and job scripts, drafting and restructuring the paper (including its title and section structure), literature search and summarization, bibliography entries, and figure creation. The estimated values marked in the paper and their stated bases were proposed with AI assistance. We have not used generative AI tools to generate, alter, or impute measured data: every reported measurement is computed by the experiment code from model runs. AI tools were not used in earlier phases of the project. The key problem this paper studies, the theoretical framework in Section 2, and the regime-map experiments were not used by AI. We have reviewed all AI-assisted work. AI-written code was checked with unit tests, including a check of the RoPE inversion against a real model’s cache, and with end-to-end runs before any GPU campaign; reported statistics were recomputed from the raw result files by the same readers that generate the tables; literature summaries were checked against the cited papers, and bibliography entries against arXiv or publisher pages. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work measures compression of the key-value cache of publicly released language models on synthetic retrieval tasks built from public-domain books (PG-19). It involves no human subjects, no personal or sensitive data, and no new model training. The experiments used GPU time on two shared academic clusters (Appendix E.4). To guard research integrity, each experiment’s decision rules were written before its results were read, negative outcomes are reported alongside positive ones, and estimated values are distinguished from measurements throughout.

REPRODUCIBILITY STATEMENT

Section 2 and Appendices C–D give the allocation problem, its approximations, and the discrete optimizer. Section 3 and Appendix E specify models, context lengths, budgets, information contracts, and the response statistics; Appendix H specifies the end-task protocol, prompt blocks, validity filter, budget and side-information accounting, and every baseline with its deviations from the original paper. The anonymized supplementary code contains the measurement and end-task runners, the baseline implementations with their unit tests, the pre-registered plans and decision rules for each experiment, the readers that produce every table and figure from result files, and the job scripts with the exact arguments used. All models are public checkpoints, the corpus is public, and prompt indices and random seeds are fixed and recorded with each result. Appendix E.4 lists the hardware, software versions, and per-job resources.

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J Extended related work

23

A SOME BASICS

The tasks are synthetic retrieval (RULER): find a random number planted somewhere in 8K–128K tokens of text, with distractor numbers nearby. There are 20 prompts per cell.

- **Synthetic Retrieval:** This means the test is artificially constructed rather than being a natural real-world task (like summarizing a real book). A piece of data is intentionally hidden inside a massive wall of text just to see if the AI can find it.
- **RULER:** This is the specific name of the benchmark framework (developed by researchers to evaluate long-context LLMs). It stands for Retrieval, Understanding, and Long-context Evaluation Robustness.
- **8K-128K tokens:** This indicates the length of the text context that the model has to process. A token is roughly a word or part of a word, so 8K tokens is about 8,000 words, and 128K tokens is about 128,000 words. The model has to search through this large amount of text to find the hidden number.
- **Distractor numbers:** These are other numbers placed in the text that are not the target number. They are meant to confuse the model and make the task harder. The model has to correctly identify the planted number and ignore the distractors.
- **20 prompts per cell: The Matrix (Cells):** When testing LLMs, researchers usually create a grid or matrix. For example, the horizontal axis might be the context length (8K, 16K, 32K, 64K, 128K) and the vertical axis might be the depth of where the needle is hidden (0% depth = top of the document, 50% = middle, 100% = bottom). Each intersection on this grid is a "cell." This means that for each configuration or "cell" of the experiment (which could be a specific model, context length, or other parameter), there are 20 different test cases (prompts) that the model is evaluated on. Each prompt is a separate instance of the retrieval task.

Some metrics

- valid mean task score

10 prompts -> 100prompts or 1000 prompts? not head-output error but task score. The task score is the final evaluation metric that measures how well the model performed on the retrieval task. It is calculated based on whether the model correctly identified the planted number in each prompt. The valid mean task score is the average score across all valid prompts (those that were not filtered out due to errors or other issues). This metric gives an overall sense of the model's performance on the retrieval task across different configurations and contexts.

B SCOPE OF THE COMMON RATE-ALLOCATION VIEW

This appendix expands the derivation and measurement contract used in the main paper. The common action space is deliberately narrow: it allocates *key-token* rates within an attention head, leaves values exact, and measures attention-output error. Prior work already combines quantization and eviction in deployable systems. Our claim is instead that a common local surrogate makes it possible to ask a different empirical question: when does the discrete interior improve over uniform quantization and eviction under matched information and key-bit budgets?

Throughout, *zero rate* means a zero-bit action in the separable planning surrogate. It does not mean that set eviction is exactly identical to independent additive logit noise. Section C makes this distinction explicit.

C OUTPUT DISTORTION AND THE ZERO-RATE APPROXIMATION

C.1 ATTENTION AND FINITE-BIT KEY PERTURBATIONS

For a query $q \in \mathbb{R}^d$, keys $k_i \in \mathbb{R}^d$, and values $v_i \in \mathbb{R}^{d_v}$, define

$$s_i = q^\top k_i / \sqrt{d}, \quad a_i = \frac{\exp(s_i)}{\sum_j \exp(s_j)}, \quad o = \sum_i a_i v_i. \quad (6)$$

Quantizing key i changes its logit by δ_i . Because

$$\frac{\partial o}{\partial s_i} = a_i(v_i - o),$$

the first-order output perturbation is

$$\Delta o \approx \sum_i a_i(v_i - o)\delta_i. \quad (7)$$

Consequently,

$$\begin{aligned} \mathbb{E}\|\Delta o\|^2 &\approx \sum_i a_i^2 \|v_i - o\|^2 \text{Var}(\delta_i) \\ &\quad + \sum_{i \neq j} a_i a_j (v_i - o)^\top (v_j - o) \text{Cov}(\delta_i, \delta_j). \end{aligned} \quad (8)$$

Equation equation 1 follows only when the residual perturbations are zero mean and the cross-token covariance contribution vanishes. Softmax invariance removes a *common* shift, $\text{softmax}(s + c\mathbf{1}) = \text{softmax}(s)$; it does not remove arbitrary token-dependent bias.

C.2 EXACT SET EVICTION

Let E be an evicted token set and $A_E = \sum_{i \in E} a_i$. Renormalizing the remaining attention weights yields

$$o_{\setminus E} = \frac{o - \sum_{i \in E} a_i v_i}{1 - A_E}, \quad (9)$$

$$o_{\setminus E} - o = -\frac{\sum_{i \in E} a_i (v_i - o)}{1 - A_E}, \quad (10)$$

$$\|o_{\setminus E} - o\|^2 = \frac{\|\sum_{i \in E} a_i (v_i - o)\|^2}{(1 - A_E)^2}. \quad (11)$$

The denominator and cross-token inner products prevent an exact token-separable representation. For one evicted token j , the exact error magnitude is

$$\frac{a_j}{1 - a_j} \|v_j - o\|,$$

which connects directly to the CAOTE singleton score. ReST-KV further models attention redistribution and output reconstruction; our local planner does not subsume those effects.

C.3 THE PLANNING APPROXIMATION

When the evicted mass is small and cross terms are neglected, Eq. equation 11 motivates

$$\tilde{D}_0(E) = \sum_{i \in E} w_i, \quad w_i = a_i^2 \|v_i - o\|^2.$$

We choose units in which the absolute zero-rate distortion is $d_0 = 1$ and write the finite-rate costs as $d_b = \text{Var}(\delta_b)$ in the same logit units. Thus

$$\tilde{D}(\mathbf{b}) = \sum_i w_i d_{b_i}, \quad b_i \in \mathcal{B}, \quad 0 \in \mathcal{B}.$$

This convention keeps relative quantizer factors separate from the absolute zero-rate cost. It is a local approximation used to choose an allocation; every reported response is obtained by recomputing attention with the actual quantized/evicted cache.

D DISCRETE RATE ALLOCATION AND REGIME COORDINATES

D.1 THE IMPLEMENTED OPTIMIZER

At average key budget B , the discrete problem is

$$\min_{b_i \in \mathcal{B}} \sum_i w_i d_{b_i} \quad \text{subject to} \quad \sum_i b_i \leq BL. \quad (12)$$

Introducing multiplier $\lambda \geq 0$ separates the token decisions:

$$b_i^*(\lambda) = \arg \min_{b \in \mathcal{B}} \{w_i d_b + \lambda b\}. \quad (13)$$

We search λ to satisfy the budget and use the discrete tiers in Eq. equation 13; the continuous expression is explanatory, not the experimental optimizer.

For the relaxation $d_b = \kappa 4^{-b}$,

$$b_i^* = \frac{1}{2} \log_2 w_i + c = \log_2 a_i + \log_2 \|v_i - o\| + c.$$

The measured 0.035-bit result is the maximum observed change in an *aggregate ladder-width statistic* after adding the value term. It is not a uniform per-token bound and does not establish that attention-only allocation is optimal. A direct value-aware versus attention-only allocation ablation is therefore retained as an explicit evidence anchor.

D.2 TIER EXTINCTION

If $d_b \geq d_0$, tier b consumes positive bits without improving the local surrogate relative to zero rate and is dominated. This is a sufficient extinction condition. The converse is not sufficient: a tier with $d_b < d_0$ can still lie above the lower convex envelope of the other rate–distortion points. For the 2-bit tier we report

$$D_2 = \frac{1}{H} \sum_{h=1}^H \mathbb{I}[d_{h,2} \geq d_0].$$

D_2 is measured at a particular model and context. Although L is absent from its formula, the head cost distribution changes with L , so D_2 is not sequence-length invariant and a new deployment context requires calibration.

D.3 WHY THE REGIME MAP NEEDS TWO COORDINATES

The sharp side and diffuse side arise for different reasons:

1. On the sharp side, large absolute finite-bit noise removes low-bit tiers; D_2 directly probes this boundary.
2. On the diffuse side, unequal allocation has little leverage when token importances are nearly equal; an importance-dispersion statistic is needed.

For attention-only importance, $\text{std}_i(\log_2 a_i) = \tau / \ln 2$, because softmax subtracts the same log normalizer from every logit. This identity alone neither proves that τ is a complete regime variable nor establishes a uniform-quantization optimum. The paper therefore treats the mixed-to-eviction side as the well-supported boundary and leaves a prospective comparison of dispersion predictors as R10.

E MEASUREMENT PROTOCOL AND ACCOUNTING

E.1 GRID AND INDEPENDENT UNITS

The symmetric R3 grid contains 16 model–context cells from six models and 29,312 query-head–configuration aggregates. The R6 boundary extension contains 25 cells spanning 4k–256k and 44,416

such aggregates. Across the six models, 10,240 query-head positions and 2,016 KV-head positions are counted once; contexts reuse those positions. Model/cell-level analyses, rather than raw heads, are therefore the independent statistical units.

PG-19 passages supply natural context. Each cell is admitted only after the uncompressed model passes the needle-retrieval validity gate. The allocation planner uses its local surrogate, while every reported error comes from actual key quantization or eviction followed by exact attention-output recomputation.

E.2 MATCHED INFORMATION AND BUDGET

The primary comparison uses $B = 3$ key bits per context token. Uniform quantization stores every key at 3 bits. The eviction endpoint retains the number of 8-bit keys allowed by the same key-bit budget. In the symmetric contract, the mixed allocator and accumulated-attention eviction both use lagged importance information; an oracle/current-query result is labeled only as an upper bound. The five-corner robustness result changes the reference set and is reported separately.

This accounting omits value-cache bytes, quantization scales, indices, metadata, protected windows, alignment, and packing. Moreover, *8-bit* is the maximum experimental key tier, not full floating-point precision. The experiments therefore establish matched key-bit output error, not matched total memory or throughput.

E.3 RESPONSES

For each head,

$$g_h = \frac{\min(D_h^{\text{uniform}}, D_h^{\text{evict}})}{D_h^{\text{mixed}}}, \quad G_+ = \exp\left(\frac{1}{H} \sum_h \log \max(g_h, 1)\right).$$

G_+ is a hindsight portfolio statistic: it clips losses after observing all three errors. It is not the performance of a deployed router. The primary cell-level response is continuous G_+ ; the fraction with $g_h \geq 2$ is a secondary thresholded band statistic. Every band result names its corner and information contracts.

E.4 COMPUTE RESOURCES AND SOFTWARE

All experiments ran as Slurm jobs on one shared academic GPU clusters. Every job recorded its GPU and library versions, and the cluster has NVIDIA H100 80GB HBM3 GPUs, PyTorch 2.13.0, and Hugging Face Transformers 5.16.1 (Python 3.11). Models were loaded in bfloat16 from pinned local checkpoints with the hub offline.

Table 4: Hardware per experiment. Each job ran on one node; wall times are measured from job records where available.

Experiment	GPUs per job	Wall time per job
Regime map, 7–8B and MoE models (§4)	1×H100	minutes to a few hours per cell
Regime map, Llama-3.3-70B and Qwen3-30B-A3B	4×H100 (70B), up to 4×H100 (30B)	up to several hours per cell
End-task audit, one model–context cell of 20 prompts (§5)	1×H100	0.5 h (8k) to 2.7 h (128k)
Router calibration, 10 prompts	1×H100	about 10 minutes
Nested-code overhead (App. H)	1×H100	1.8–2.8 h per cell
Readers, statistics, figures, tests	CPU only	minutes

Because the model forward pass is not bit-reproducible across processes even on one GPU, every comparison that pairs methods prompt by prompt computes all methods inside one process on the same captured queries, keys, and values; comparisons across processes are made on cell means only.

Table 5: Measured costs of acting on the output-distortion map. Output error only.

Constraint	Measurement	Implication
Family label over 4,096 tokens	p90 regret 1.00 in 6/6 cells	calibrate the route once per context
Frozen token assignment	$2.52\text{--}4.30 \times$ lag-1 error	re-budget mixed heads every few steps
Shared GQA allocation	$1.000/1.136/1.316 \times$ at $n_{\text{rep}} = 1/4/8$	route per KV head at a small cost
4-bit current-query cascade	closes 82–91% (median 89%) of lag penalty	re-score from base-tier keys

F ADDITIONAL ROBUSTNESS AND DEPLOYMENT EVIDENCE

The completed measurements support the following bounded conclusions:

- In all 16 R3 cells, the symmetric lagged-versus-lagged band exceeds the all-oracle band by 1.2–12.5 points, but it is lower than the asymmetric current-query-versus-lagged estimate. Five cells are STOP and five NARROW.
- Across the 25 R6 cells, D_2 has Spearman correlation -0.978 with both G_+ and the band; the model-partial correlation is -0.915 . Similar D_2 values can nevertheless differ by eight band points across architectures.
- Prompt-block standard deviation is 0.4–3.0 band points. Replacing the two headline corners by the five-corner minimum lowers the band by 5.1–8.6 points and can change a categorical verdict.
- In six R5 cells, the calibration-time family choice has p90 regret 1.00 through 4,096 generated tokens, whereas a frozen token allocation costs $2.52\text{--}4.30 \times$ a one-step-old allocation.
- Enforcing one allocation per shared KV head gives marginal in-band costs 1.000, 1.136, and 1.316 at GQA repetition factors 1, 4, and 8. A 4-bit current-query cascade closes 82–91% (median 89%) of the lagged-score penalty.

Table 6: Predictor comparison on the 25 regime-map cells (§4). Columns: pooled Spearman correlation with G_+ ; the same partialled on model identity; leave-one-model-out error of a log-linear fit of G_+ , as a multiplicative factor; mean Spearman within the held-out model. All predictors are cell medians of per-head medians on the rows that define D_2 ; the recomputed D_2 matches the reported one to within 0.52 points in every cell. The heterogeneity statistic is the spread of $\frac{1}{2} \log_2 w_i$; for log-normal w it ranks cells as the AM/GM ratio does.

predictor	ρ with G_+	ρ model	held-out model error	within-model ρ
D_2 (ours)	-0.978	-0.942	$1.119 \times$	-0.986
spread of log sensitivity (heterogeneity)	-0.722	-0.923	$1.155 \times$	-0.986
attention concentration τ	-0.724	-0.922	$1.154 \times$	-0.986
attention entropy	-0.288	-0.903	$1.273 \times$	-0.943
n_{95}/L	$+0.331$	$+0.021$	$1.271 \times$	$+0.286$
context length only	-0.478	-0.872	$1.243 \times$	-0.986

Methods notes and negative results. Three further results inform the protocol but are not findings. (i) *Fitting the crossing.* A logistic fit of band against D_2 places the GO crossing at 34.6%, although no cell lies between 25.8% and 27.0%, where the band actually crosses 35%; the flat tail above 30% pulls the curve. We therefore report isotonic (monotone) crossings, which cannot leave the bracket of measured cells, and use the logistic only as a flagged cross-check. (ii) *Per-head budgets under GQA.* Measured per query head, the smallest keep count within 10% of the full-budget error (K^*) averages 0.72–0.98 of the budget at 32k. Taking the maximum over the query heads that share a KV head, as a GQA cache must, raises this above 0.98 in four of five GQA models (0.92 in Qwen3-8B), while the MHA control is unchanged at 0.80. Head-level slack largely disappears at the granularity a cache can store; this agrees with the per-KV-head costs in Table 5. (iii) *Own-baseline budget rules.* A direct integer scan of physical-group K^* at 8k is stable across prompts (Spearman 0.75) but nearly constant

(coefficient of variation 0.006), so a K^* -proportional budget moves only 0.23% of memory. Worse, K^* is anti-correlated with the absolute full-budget error (Spearman -0.80): the groups it would take budget from hold 85% of the error mass. Any rule that sizes each unit by a tolerance relative to its *own* baseline error therefore allocates away from the units that dominate a global objective. This qualification used two models and two held-out prompts and stopped at its pre-registered gate; we report it as a negative result, not a characterization.

One predictor study remains intentionally unfilled. R10 must compare D_2 prospectively against RateQuant-style heterogeneity, entropy/Gini, and context-only predictors under a fixed leave-model-out protocol. The end-task audit is reported in Appendix H. Beyond it, two natural-task (LongBench-v2 (Bai et al., 2025) forced-choice) development blocks expose only 3/52 and 2/52 candidate headroom over their best fixed policies; sequential gates suppress both proxy outcomes and leave confirmation unopened.

G CLAIM BOUNDARY

Supported	Not yet supported
Zero rate is useful inside a stated separable local surrogate	Exact equivalence between independent quantization noise and set eviction
An absolute tier cost can make a finite-bit tier dominated	A universal one-dimensional phase transition or threshold
D_2 strongly orders the measured mixed-to-eviction boundary	A context-free predictor that transfers without calibration
The output-error-optimal family label is temporally stable in six measured cells while token allocations become stale quickly	A robust end-task router, total-KV memory savings, or realized throughput
The output-error-optimal allocator is the strongest token-selective policy on the measured retrieval cells	That it beats dense low-bit quantization in general (it loses on Llama-3.1-8B)
Output distortion orders budgets and contexts by difficulty	That output distortion can choose between method families at a fixed budget
Token-granular allocation truncates multi-token answers	That positional smoothing repairs it (pre-registered test pending)

Table 7: The empirical and methodological claim boundary of the current draft.

H END-TASK AUDIT: PROTOCOL, BASELINES, AND FULL RESULTS

Protocol. Each prompt’s context is prefilled at full precision and scored; the cache is then compressed once, and the question is prefilled *through* the compressed cache before greedy decoding. This is the question-agnostic setting of a prefix cache or multi-turn session. When the question is inside the observation window instead, SnapKV scores 1.00 at every budget, because the window vote points at the needle. Evaluation uses prompts 100–119; the SIEVE router is calibrated on the disjoint prompts 0–9. A block is valid when the uncompressed model scores at least 0.9; this excludes Qwen3-8B multi-value retrieval (0.24 at 8k, 0.04 at 32k). Scores are RULER string matches; multi-value and variable tracking give fractional credit.

Budget accounting. Every method spends B key code bits per context-key element, verified at run time from the applied bit maps. Side information differs by method and is reported in Table 8 rather than folded into B . Eviction and SIEVE keep tokens at 8 bits with the same TurboQuant quantizer, so their side information scales with the kept fraction.

Storage costs outside the code budget. Two further costs are small relative to the differences reported in the main text. *Nested tiers.* A mixed allocator must store tokens at several widths. We measured the rate cost of storing the tier ladder $\{0, 3, 4, 6, 8\}$ as one successively refinable $3+1+2+2$

code, in which an 8-bit index truncates to 6, 4, or 3 bits without re-encoding, against separate Lloyd–Max codebooks, at matched attention-output error; successive refinability is classical (Equitz & Cover, 1991), but our scalar codes are not the asymptotic optimum, so we measured it. The code itself costs at most 0.42% rate at any width. End to end at $B = 3$, including the extra bit that 3-bit tokens keep for the 4-bit cascade score, the overhead is +0.34% averaged over four model–context cells (90% CI $[-0.33, +0.99]$) and +2.9% in the worst cell. *Protected window*. The last 32 context tokens stay uncompressed in every method: 0.39% of the context at 8k, 0.10% at 32k, and 0.02% at 128k. Because it is identical across methods, it shifts every method’s memory equally and does not affect the comparison.

Table 8: Baselines as implemented, and side information per key element ($d = 128$).

method	what is stored, and how it is chosen	side bits
TurboQuant-MSE	every token at B bits; random rotation, Lloyd–Max, per-token norm	0.125
KIVI ($g = 32 / 128$)	every token at B bits; per-channel asymmetric integer, min/max per g tokens, post-RoPE	1.0 / 0.25
KVQuant	every token at B bits; pre-RoPE per-channel non-uniform codebook; 1% outliers and token 0 exact	≈ 0.32
H2O	$\lfloor BC/8 \rfloor$ tokens at 8 bits; attention summed over all prefill queries	≈ 0.05
SnapKV	same keep count; 32-query observation window, max-pool 7	≈ 0.05
DropKV, Ada-KV, OBCache-K, LaProx SIEVE	same keep count; published scores; Ada-KV head allocation for Ada-KV and OBCache-K; LaProx global top- K per-token width in $\{0, 1, 2, 3, 4, 5, 6, 8\}$; Eq. 3 per KV head, 4-bit cascade score; offline per-KV-head router	≈ 0.05 0.07–0.09

Deviations from the original papers. KIVI keeps its last partial group in full precision; we quantize it as a short group, and the 32-token uncompressed window shared by all methods plays the role of KIVI’s residual. KVQuant calibrates outlier thresholds and its non-uniform codebook offline with Fisher-weighted k -means; we fit both online on each prompt’s own context without weighting, which can only favor it. H2O uses its raw accumulated score, whose causal position bias (a token is seen by every later query) is H2O’s own. All methods use simulated quantization: the model reads dequantized keys, and no packed kernel is involved.

Table 9: SIEVE (router) against each baseline over the 36 valid cells. A win or loss is a cell-mean difference above 0.05; the interval resamples prompts within each model and context.

SIEVE (router) vs.	wins	ties	losses	mean Δ	90% CI
TurboQuant-MSE	5	13	18	−0.180	$[-0.208, -0.152]$
SnapKV	31	1	4	+0.367	$[+0.338, +0.398]$
DropKV	30	3	3	+0.348	$[+0.314, +0.384]$
Ada-KV	23	3	10	+0.262	$[+0.236, +0.290]$
LaProx	22	4	10	+0.187	$[+0.154, +0.221]$
OBCache-K + Ada-KV	21	0	15	+0.163	$[+0.129, +0.197]$
best eviction method per cell	21	0	15	+0.127	—

I ESTIMATE LEDGER

Numbers typeset as $x^\dagger$ in the main text are estimates, not measurements. Each is replaced by the run named below; the paired rerun (R12-A) also replaces every measured end-task number, because the forward pass is not bit-reproducible across processes and all methods must share one process per cell to be compared prompt by prompt.

Table 10: Failure case: Llama-3.1-8B at 128k. Error types are over single- and multi-key answers at $B \in \{2, 3\}$ (80 per method). *Truncated*: the prediction is a proper prefix of at least three digits of the correct number. *Wrong needle*: another needle’s value.

method	task score		single/multi-key answers, $B=2, 3$			evicted ($B=2$)	head-output error ($B=2$)	heads to interior
	$B=2$	$B=3$	truncated	wrong needle	other			
TurboQuant-MSE	0.942	0.982	0%	1%	0%	0%	0.706	—
SnapKV	0.523	0.652	6%	18%	11%	75%	0.187	—
DropKV	0.480	0.657	11%	16%	10%	75%	0.201	—
Ada-KV	0.704	0.872	1%	15%	1%	75%	0.181	—
LaProx	0.737	0.863	5%	12%	1%	75%	0.187	—
OBCache-K + Ada-KV	0.762	0.902	4%	10%	1%	75%	0.182	—
SIEVE interior only	0.409	0.622	39%	6%	6%	65%	0.096	—
SIEVE (router)	0.442	0.677	36%	4%	9%	64%	0.094	99%

Table 11: Full end-task grid: mean score over 20 prompts per cell. Bold marks the best method in each valid row. ‡: uncompressed score below 0.9, excluded from every aggregate. KIVI, KVQuant and H2O are added by the paired rerun (§I).

model	ctx	task	B	FP	TurboQ.	SnapKV	DropKV	Ada-KV	LaProx	OBC+Ada	SIEVE-int.	SIEVE
Llama-3.1-8B	8K	multi-key	2	1.00	1.00	0.20	0.30	0.25	0.25	0.40	0.35	0.30
Llama-3.1-8B	8K	multi-key	3	1.00	1.00	0.35	0.55	0.55	0.55	0.75	0.60	1.00
Llama-3.1-8B	8K	multi-value	2	1.00	1.00	0.14	0.16	0.19	0.25	0.39	0.15	0.23
Llama-3.1-8B	8K	multi-value	3	1.00	1.00	0.34	0.42	0.42	0.46	0.59	0.40	0.99
Llama-3.1-8B	8K	single	2	1.00	1.00	0.45	0.55	0.50	0.60	0.70	0.30	0.35
Llama-3.1-8B	8K	single	3	1.00	1.00	0.75	0.65	0.75	0.80	0.75	0.60	1.00
Llama-3.1-8B	8K	var. track.	2	1.00	0.98	0.45	0.49	0.42	0.40	0.54	0.43	0.60
Llama-3.1-8B	8K	var. track.	3	1.00	1.00	0.60	0.64	0.62	0.59	0.88	0.67	0.95
Llama-3.1-8B	32K	multi-key	2	1.00	1.00	0.30	0.20	0.35	0.45	0.55	0.40	0.45
Llama-3.1-8B	32K	multi-key	3	1.00	1.00	0.45	0.35	0.70	0.70	0.70	0.60	1.00
Llama-3.1-8B	32K	multi-value	2	1.00	0.97	0.12	0.06	0.20	0.26	0.35	0.16	0.23
Llama-3.1-8B	32K	multi-value	3	1.00	1.00	0.42	0.23	0.56	0.61	0.70	0.40	0.90
Llama-3.1-8B	32K	single	2	1.00	1.00	0.60	0.35	0.65	0.75	0.85	0.55	0.60
Llama-3.1-8B	32K	single	3	1.00	1.00	0.70	0.60	0.95	0.95	0.90	0.65	1.00
Llama-3.1-8B	32K	var. track.	2	1.00	0.99	0.28	0.37	0.46	0.47	0.69	0.46	0.63
Llama-3.1-8B	32K	var. track.	3	1.00	1.00	0.65	0.55	0.82	0.87	0.86	0.64	1.00
Llama-3.1-8B	128K	multi-key	2	1.00	1.00	0.30	0.40	0.50	0.60	0.65	0.35	0.40
Llama-3.1-8B	128K	multi-key	3	1.00	0.95	0.45	0.50	0.80	0.70	0.85	0.50	0.55
Llama-3.1-8B	128K	multi-value	2	0.97	0.94	0.41	0.35	0.57	0.59	0.64	0.34	0.34
Llama-3.1-8B	128K	multi-value	3	0.97	0.99	0.54	0.55	0.79	0.81	0.85	0.55	0.69
Llama-3.1-8B	128K	single	2	1.00	1.00	0.90	0.75	1.00	0.95	0.95	0.45	0.45
Llama-3.1-8B	128K	single	3	1.00	1.00	0.95	0.85	1.00	1.00	0.95	0.65	0.65
Llama-3.1-8B	128K	var. track.	2	1.00	0.83	0.48	0.42	0.74	0.81	0.81	0.50	0.58
Llama-3.1-8B	128K	var. track.	3	1.00	0.99	0.67	0.73	0.90	0.94	0.96	0.79	0.82
Qwen3-8B	8K	multi-key	2	1.00	0.95	0.00	0.05	0.00	0.05	0.05	0.15	0.90
Qwen3-8B	8K	multi-key	3	1.00	1.00	0.10	0.20	0.15	0.30	0.10	0.50	1.00
Qwen3-8B	8K	multi-value [‡]	2	0.24	0.69	0.00	0.05	0.00	0.01	0.01	0.05	0.72
Qwen3-8B	8K	multi-value [‡]	3	0.24	0.49	0.00	0.07	0.00	0.21	0.10	0.28	0.59
Qwen3-8B	8K	single	2	1.00	0.95	0.10	0.10	0.05	0.25	0.15	0.20	1.00
Qwen3-8B	8K	single	3	1.00	1.00	0.25	0.35	0.25	0.55	0.40	0.60	1.00
Qwen3-8B	8K	var. track.	2	1.00	0.46	0.19	0.28	0.28	0.41	0.43	0.47	0.99
Qwen3-8B	8K	var. track.	3	1.00	1.00	0.63	0.63	0.62	0.66	0.68	0.81	1.00
Qwen3-8B	32K	multi-key	2	1.00	0.75	0.00	0.15	0.05	0.15	0.15	0.10	1.00
Qwen3-8B	32K	multi-key	3	1.00	1.00	0.10	0.30	0.10	0.30	0.25	0.40	1.00
Qwen3-8B	32K	multi-value [‡]	2	0.04	0.75	0.00	0.00	0.00	0.05	0.00	0.01	0.12
Qwen3-8B	32K	multi-value [‡]	3	0.04	0.42	0.00	0.04	0.00	0.03	0.00	0.00	0.07
Qwen3-8B	32K	single	2	1.00	0.95	0.20	0.30	0.20	0.40	0.25	0.35	1.00
Qwen3-8B	32K	single	3	1.00	1.00	0.30	0.55	0.50	0.70	0.55	0.45	1.00
Qwen3-8B	32K	var. track.	2	1.00	0.40	0.28	0.40	0.45	0.79	0.61	0.46	0.99
Qwen3-8B	32K	var. track.	3	1.00	0.95	0.69	0.70	0.77	0.92	0.81	0.76	0.99

J EXTENDED RELATED WORK

Low-precision formats beyond the KV cache. Block-scaled FP4 has moved from storage compression to full-stack inference and training. NVFP4 pairs 4-bit E2M1 values with FP8 scales over 16-element blocks (Alvarez et al., 2025a; NVIDIA, 2025b), finer than the power-of-two, 32-element scales of MXFP4 (Open Compute Project, 2023), and has been used to pretrain LLMs (NVIDIA, 2025a; Devleker & Ghodsian, 2025). Libraries provide fused FP8/FP4 kernels and quantization recipes (NVIDIA, 2024b;a). Open-weight models ship with FP4 expert weights (OpenAI, 2025;

quantity	estimate	basis	replaced by
H2O, full grid, Llama / Qwen	0.38 / 0.33	Measured in the 8k rerun at 0.354 / 0.358; question-aware runs at 32k gave 0.37 ($B = 2$) and 0.48 ($B = 3$).	R12-A
KIVI ($g=128 / g=32$), full grid	0.96 / 0.98 (all)	Measured in the 8k rerun: Llama 0.964 / 0.986, Qwen 0.987 / 1.000. On Qwen3-1.7B at 2k TurboQuant scored 0/11 at both budgets while KIVI and KVQuant held.	R12-A
KVQuant, full grid	0.97 (all)	Measured in the 8k rerun: Llama 0.976, Qwen 0.990.	R12-A
Hard cell, all methods	see Table 2	Screening block (prompts 500–539): FP 1.00, TurboQuant 0.825. On the 10 calibration prompts (1100–1109) FP and TurboQuant both score 0.90, so the pre-registered gate may fail; other methods scaled from 4-key retrieval at 32k.	R12-B
Mistral-7B, SIEVE router	0.55	Calibration prompts 0–9: FP fails variable tracking (0.2) at both contexts and multi-value at 32k; on valid tasks TurboQuant 0.93–1.00, SnapKV 0.03–0.10, SIEVE interior 0.00–0.27.	R12-D
Pooled-score interior, 128k, $B = 2$	0.58	At 8k it moved the Llama interior from 0.299 to 0.663 at $B = 2$ and truncation from 39% to 9%; 128k is the pre-registered cell.	R12-A (diag.)
Oracle router, 128k, $B = 2$	0.55	At 8k the oracle adds +0.15 on Llama (0.680 to 0.834); scaled to 128k, where the calibrated router sends 99% of heads to the interior.	R12-A (diag.)

Table 12: Closest work and the claim boundary of this paper.

Work	Allocation/action	Difference here
DiffKV; HqeKV	token high/low precision and pruning	explain and predict when the middle tier pays; no first-hybrid claim
CAOTE; CriticalKV; DropKV; OBCache; LaProx; ReST-KV	output-aware token eviction	finite-rate extension under a stated local surrogate; exact set error retained; evaluated as baselines where available
RateQuant	K/V heads, finite rates	within-head key tokens, explicit zero rate, context traversal
RDKV (concurrent)	value tokens, key channels, zero-to-finite rates	when the token-wise interior beats endpoints; score staleness and regime movement
AATC (concurrent)	transformed K/V channels, including zero	token-axis tier viability; values/channels remain outside our allocation
Fixed-contract diagnostic	attribution of eviction failures	measures how comparison set and information contract move verdicts; truncation mechanism

DeepSeek-AI, 2026), FP4 has been applied to diffusion models by absorbing outliers into low-rank branches (Li et al., 2025) and to diffusion transformers for image and video (Zhao et al., 2024), and NVFP4 backbones have been combined with LoRA for reinforcement learning (Huang et al., 2025). For our question, these formats strengthen the dense endpoint: a 4-bit cache with hardware support is the baseline any mixed or evicting policy must beat. Its 16-element scales also limit how far a few large key channels can inflate the quantization step, the weakness of per-token scaling that we suspect behind TurboQuant’s failure on Qwen3 (F4); we have not tested NVFP4 caches here.

Low-precision attention kernels. FlashAttention-2 and -3 fuse attention without materializing the score matrix, the latter with FP8 (Dao, 2023; Shah et al., 2024); SageAttention quantizes the attention computation itself to 8 bits, INT4 with outlier smoothing, and microscaled FP4 (Zhang et al., 2024b;a; 2025a). Because these kernels never write attention scores to memory, evictors that read historical scores (H2O’s accumulated attention, and the lagged scores our allocator uses) must either fall back to eager attention or estimate importance another way, for example from pre-RoPE geometry (Mao et al., 2026). Our experiments use an eager attention path and therefore measure accuracy, not this cost.

KV caches in video generation. Autoregressive video generators accumulate caches across frames. Quant VideoGen quantizes them to 2 bits using the redundancy of adjacent frames (Xi et al., 2026); LongLive supports real-time interactive long video (Yang et al., 2026), and LongLive 2.0 builds an NVFP4 parallel inference stack with fused dequantization (Chen et al., 2026) on open video generators (Wan Team, 2025). The strong temporal redundancy in video changes the regime question: when adjacent tokens are near duplicates, token selection and low precision interact differently than in the text retrieval settings studied here.